

Shot-making or shot selection? Separating the two in NBA shooting

An expected-make model validated on unseen seasons, then three questions for a front office: how much shooters differ at making shots, whether it lasts, and what teams control.

Andres Perez, M.S. Kinesiology (Biomechanics) Python · scikit-learn · statsmodels

Code: github.com/Andresperez397/nba-shot-making

DATA

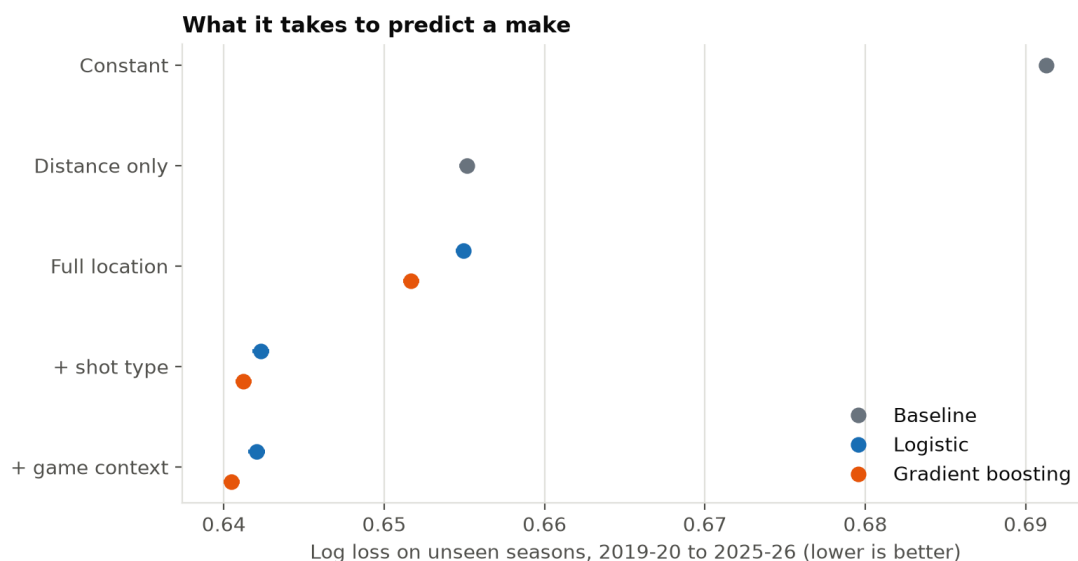
1,897,373 regular-season field-goal attempts from 2017-18 to 2025-26 (NBA Stats play-by-play). Inputs are shot location, shot type and game context. The model is tested on seven seasons it never trained on (1.47M shots).

APPROACH

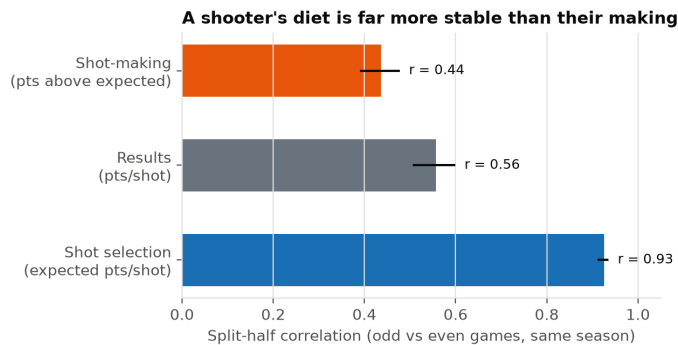
- **Audit, then a frozen plan:** fields that reveal the result (description text, assists, blocks, the shot row's own score) are excluded; the pre-shot score is rebuilt from earlier events.
- **Out-of-sample expected makes:** each season is scored by a model trained only on earlier seasons.
- **Empirical-Bayes shrinkage** for player shot-making; cluster bootstraps for every comparison.
- **A data break caught:** the NBA moved rim-shot coordinates in 2017-18, so the window starts there.

FINDINGS

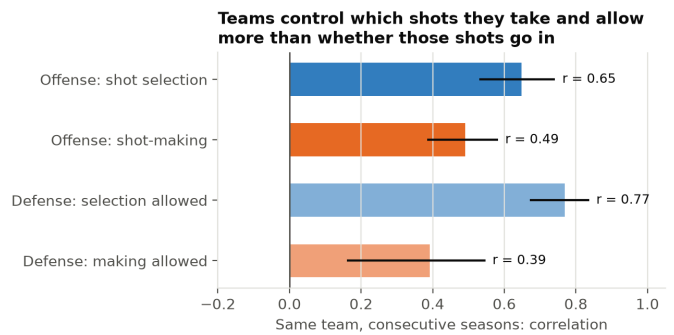
- **Location and shot type predict makes;** game context adds almost nothing. Gradient boosting beat a spline logistic model, and the model is well calibrated (slope 1.03).
- **Shooters differ by about 7 points per 100 shots** in true shot-making. Jokić holds the six best seasons.
- **Diet is far steadier than making:** split-half r is 0.93 for shot selection and 0.44 for shot-making.
- **Making is still a skill:** shrunk shot-making predicts next season's better than raw numbers or assuming no skill ($r = 0.55$).
- **Defenses control the shots they allow** (year-to-year r 0.77) far more than whether opponents make them (0.39).



Held-out log loss by information layer, pooled over 2019-20 to 2025-26, with 95% game-bootstrap intervals (narrower than the dots).

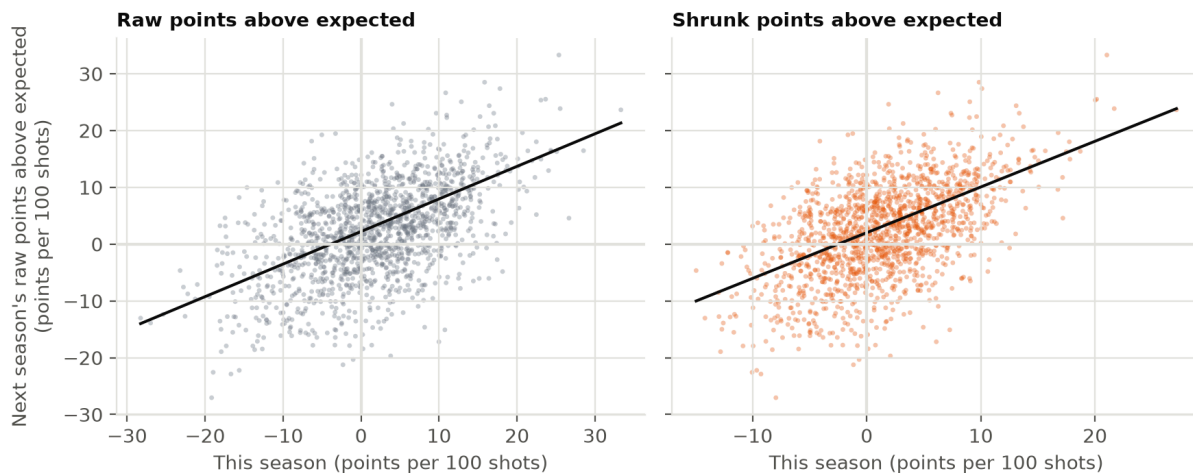


Odd vs even games within a season, 2,305 player-seasons with at least 200 attempts. 95% player-bootstrap intervals.



Same team in consecutive seasons, 180 pairs. Defensive selection-allowed exceeds making-allowed by 0.38 (95% CI 0.21–0.60).

Shot-making carries over to next season, and shrinkage makes it carry over better



1,545 consecutive player-season pairs. Shrunk shot-making predicts next season's better than raw (weighted MSE 0.0042 against 0.0054) and better than assuming no skill (0.0072).

ROBUSTNESS AND LIMITATIONS

Results hold with location-only expected makes, which avoid scorer-assigned shot-type labels (the two shot-making estimates correlate 0.96), and with shot-making centered within each season. Public data have no defender positions, so part of measured shot-making is shot quality the data can't see, which may favor bigs and players who get open looks. Tips and putbacks have no charted location after 2020-21.

Data: NBA Stats play-by-play via the sportsdataverse project. The data belong to the NBA; used for non-commercial research; raw data not redistributed. Not affiliated with or endorsed by the NBA.